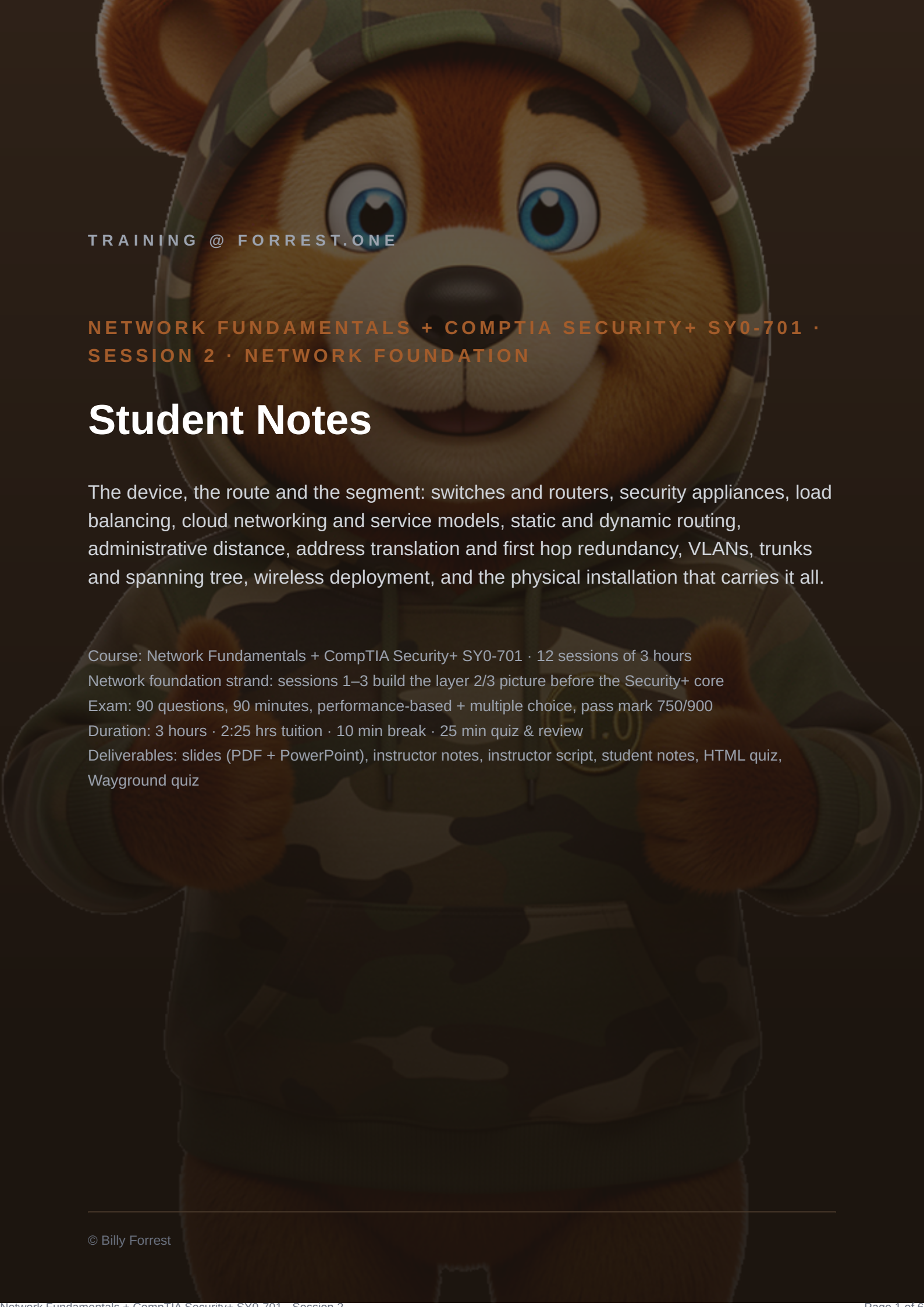




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 2 · NETWORK FOUNDATION

Student Notes

The device, the route and the segment: switches and routers, security appliances, load balancing, cloud networking and service models, static and dynamic routing, administrative distance, address translation and first hop redundancy, VLANs, trunks and spanning tree, wireless deployment, and the physical installation that carries it all.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Network foundation strand: sessions 1–3 build the layer 2/3 picture before the Security+ core

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Student Notes, Switching, Routing & Network Architecture

Your revision guide for Session 2, the second of three network-foundation sessions. Read alongside the slides, trace the frame walkthrough and the route selection order by hand, then use the self-check questions and the session quiz to confirm you are ready for the services session and the Security+ core. The running theme is that every device and every segment is a boundary, and that segmentation is an architectural control rather than tidiness.

Key terms

Term	Definition
Frame	Layer 2 PDU, Ethernet headers plus payload
Packet	Layer 3 PDU, forwarded by a router using IP addresses
MAC address	48-bit hardware address, six hex pairs; the first three octets are the OUI
Forwarding table	The switch's learned map of MAC addresses to ports
Unknown unicast flooding	Frame sent to every port except the source when the destination is unknown
MAC flooding	Overflowing the forwarding table so the switch degrades to hub mode
Firewall	Filters traffic on rules built from addresses, ports, protocols and content
IDS	Intrusion detection system: monitors passively and alerts, never blocks
IPS	Intrusion prevention system: sits inline and can drop, block or reset
Load balancer	Spreads requests across a server pool; round robin or least connections
Forward proxy	Hides the client and filters outbound requests
Reverse proxy	Fronts the servers, caches content and terminates TLS
NAS	Network attached storage: file level sharing over the network, NFS and SMB
SAN	Storage area network: block level storage, usually over Fibre Channel
NFV	Network functions virtualisation: appliances become software on ordinary servers
VPC	Virtual private cloud: a logically isolated network inside a public cloud
Security group	Stateful firewall attached to a cloud resource; replies are allowed automatically

Term	Definition
Network ACL	Stateless filter at the subnet boundary of a cloud network
Static route	A route typed by hand; administrative distance 1; it never adapts
Distance vector	A protocol where each router advertises its whole table: routing by rumour
Link state	A protocol where every router holds the same map and calculates its own paths
Path vector	A protocol that selects on attributes and policy; BGP is the internet example
Administrative distance	A ranking of how trustworthy a route source is; the lowest value wins
Longest match rule	The most specific route wins, before distance or metric is considered
NAT	Network address translation: rewrites addresses in the packet header
PAT	Port address translation: many private addresses onto one public address by port
First hop redundancy	HSRP, VRRP and GLBP: a shared virtual gateway address that survives failure
Subinterface	A logical interface on one physical router port, serving one VLAN
VLAN	A logical broadcast domain built inside the switching hardware
Trunk port	Carries many tagged VLANs between switches
Access port	Carries a single untagged VLAN to one endpoint
802.1Q tag	Four byte header carrying the 12-bit VLAN identifier
Native VLAN	The untagged VLAN on a trunk; VLAN 1 by default and a known risk
VLAN hopping	Crossing VLAN boundaries by switch spoofing or double tagging
Port security	Limits the MAC addresses allowed on a port; violations can shut it down
PortFast	Skips the listening states on a port that leads to one endpoint
BPDU Guard	Shuts a port that receives a bridge protocol data unit
Root Guard	Prevents a port from becoming a root port
STP	IEEE 802.1D, blocking redundant ports to keep one loop free path
RSTP	IEEE 802.1w, converging in seconds with alternate and backup roles
Root bridge	The switch with the lowest bridge identifier: priority plus MAC address
MTU	Maximum transmission unit; 1500 bytes on standard Ethernet
Jumbo frame	An Ethernet frame up to about 9000 bytes; every device must agree
SSID	Service set identifier, the wireless network name clients see
BSSID and ESSID	Single access point identifier, and the shared name that allows roaming

Term	Definition
WPA2 and WPA3	Wireless encryption with AES; WPA3 adds simultaneous authentication of equals
802.1X	Authenticates a device before it joins the network, using a central server
DMZ	Segment hosting public facing services, isolated from the internal network
Blast radius	How far a compromise can reach; segmentation reduces it

Core concepts

Chapter 1: Infrastructure devices

A switch forwards frames at layer 2 by MAC address, building its table from source addresses and flooding unknown destinations to every port except the source. A router forwards packets at layer 3 by IP address between networks. Firewalls filter on rules, intrusion detection monitors and reports, and intrusion prevention sits inline and blocks. Load balancers spread requests across a pool; a forward proxy hides clients and a reverse proxy fronts servers and terminates TLS. NAS serves files with NFS and SMB; a SAN serves blocks over Fibre Channel. Access points bridge wireless clients to the wired network, controllers manage them centrally, and wireless bridges join two segments with no cable.

Chapter 2: Cloud networking and routing foundations

Network functions virtualisation turns firewalls, routers and balancers into software, which lowers cost but moves the attack surface to the management plane and the hypervisor. A virtual private cloud is a logically isolated cloud network, with security groups as stateful per-resource firewalls and network ACLs filtering at the subnet boundary. Deployment models are public, private and hybrid; service models are software, infrastructure and platform as a service, and each one shifts, but never removes, responsibility. Routing starts with static routes, which never adapt, then extends to distance vector protocols such as RIP, where fifteen hops is the maximum and sixteen means unreachable, and link state protocols such as OSPF, where every router holds the same map of the network.

Chapter 3: Route selection and translation

A router chooses in a strict order: the longest match first, then administrative distance to pick the source it trusts, then the metric to pick between paths from that source. Connected routes score 0, static 1, EIGRP 90, OSPF 110 and RIP 120. EIGRP is advanced distance vector with a pre-calculated backup path; BGP is the path vector protocol of the internet and selects on policy. NAT rewrites addresses as traffic crosses a boundary, with static NAT for published servers and dynamic NAT from a pool; PAT multiplexes many addresses onto one by port number, which is why inbound services need port forwarding. First hop redundancy gives routers a shared virtual gateway address, so clients never notice a failure. Router on a stick gives each VLAN a subinterface and a gateway over one shared trunk, which is simple but can become the bottleneck.

Chapter 4: VLANs, switching and spanning tree

A VLAN is a logical broadcast domain, giving containment: fewer broadcasts, a smaller blast radius and no easy lateral movement. Trunks carry many VLANs between switches with an 802.1Q tag, and the native VLAN crosses untagged, so a mismatch can leak traffic between VLANs. Switch spoofing and double tagging are the two VLAN hopping attacks. Port configuration matters too: speed and duplex agree, port security limits addresses per port, PortFast skips the listening states on an endpoint port and BPDU Guard shuts a port that receives a BPDU. Redundant links create loops, so STP elects a root bridge on the lowest bridge identifier and blocks the redundant ports. Rapid spanning tree converges in seconds, multiple spanning tree maps VLAN groups to instances, and Root Guard, BPDU Guard and port security are the defences. Finally, the MTU is 1500 bytes, jumbo frames reach about 9000, and every device in the path must agree or packets vanish.

Chapter 5: Wireless and physical deployment

In the 2.4 GHz band only channels 1, 6 and 11 do not overlap; 5 GHz offers up to twenty three non overlapping channels and 6 GHz far more again. The SSID is the network name, the BSSID names one access point and the ESSID lets clients roam. Hiding the name is not a control. Encryption runs from WEP, which is broken, through WPA and WPA2 with AES, to WPA3 with simultaneous authentication of equals. Enterprise mode uses 802.1X with a central authentication server, Protected EAP inside a TLS tunnel and EAP TLS with certificates. A guest network gets its own name, its own VLAN and a bandwidth limit. Access points are placed by survey, not by eye, and the room itself matters: a 19 inch rack, 1.75 inch rack units, labelled cable, a documented patch panel, locked racks, hot and cold aisles and a UPS.

Practical cautions

- Do not confuse switches (layer 2, frames) with routers (layer 3, packets).
- Do not assume an IDS stops anything: it alerts, and only an inline IPS blocks.
- Do not mix up the proxies: forward hides clients, reverse fronts servers and terminates TLS.
- Do not treat NAS and SAN as interchangeable; the protocols and the level of access differ.
- Do not assume the cloud provider secures your data; service models shift responsibility rather than remove it.
- Do not leave the native VLAN at VLAN 1, and do not leave a trunk with an implied allow-all list.
- Do not confuse administrative distance (which source) with the metric (which path).
- Do not confuse NAT with PAT: PAT is the port-based form that hides many hosts behind one address.
- Do not confuse BPDU Guard (shuts the port) with Root Guard (stops root takeover).
- Do not assume jumbo frames help everywhere; a mismatch in the path silently drops large packets.
- Do not count a hidden SSID or a long WEP key as wireless security.
- Do not neglect physical access: it defeats every logical control you have.

Self-check questions

Answers are in your session slides

If you cannot answer a self-check from memory, revisit the corresponding slide, then retake the quiz. The twenty quiz questions are tagged by topic area, so any weak area is easy to find and revise.

Exam tips for Session 2

- Learn, look up, forward, flood: the switch behaviour sequence is a classic question.
- Know the appliance split cold: the firewall refuses by rule, detection reports, prevention blocks inline.
- Link each device to its protocol: NAS with NFS and SMB, SAN with Fibre Channel, access points with 802.11.
- For any cloud question, ask where the responsibility boundary sits before answering.
- Memorise the administrative distance table, and remember the longest match rule is applied before it.
- Remember PAT tells sessions apart by port number, and that inbound services need a forwarding rule.
- The native VLAN rule is examinable: move it, prune unused VLANs, restrict the trunk allowed list.
- Distinguish switch spoofing from double tagging whenever VLAN hopping appears.
- STP port states, RSTP convergence and the guard features appear in scenario questions; know which guard does what.
- Segmentation as a control answers most 'contain the breach' and lateral movement questions.
- Wireless answers default to WPA2 or WPA3 with AES, enterprise mode with 802.1X, and a separate guest VLAN.

Further reading

- IEEE 802.1Q, VLAN tagging and trunking.
- IEEE 802.1D, Spanning Tree Protocol; IEEE 802.1w, Rapid Spanning Tree; IEEE 802.1s, Multiple Spanning Tree.
- IEEE 802.1X, port based network access control, with the EAP methods that carry it.
- RFC 826, Address Resolution Protocol, and its security implications.
- RFC 4271, Border Gateway Protocol version 4, and route hijacking as a security concern.
- CompTIA Security+ SY0-701 Exam Objectives, Domain 3 (Security Architecture) and Domain 4 (Operations).